

PSAC Science Specimen Paper 2026 - Mark Scheme

Question/ Item	Answer	Mark	Additional guidance
1(a)	A	1	
1(b)	A	1	
1(c)	D	1	
1(d)	D	1	
1(e)	B	1	
1(f)	A	1	
1(g)	C	1	
1(h)	C	1	
1(i)	D	1	
1(j)	B	1	
Total marks		10	
2(a)	Coal: Non-renewable and polluting	1	
	Bagasse: Renewable and polluting	1	
	Water: Renewable and non-polluting	1	
2(b)	Chemical	1	
	sound	1	
	Electrical	1	
	heat	1	
Total marks		7	
3(a)(i)	C	1	
3(a)(ii)	Condensation	1	Irrespective of order
	Precipitation	1	
3(b)	The water vapour in the air condensed when they touched the cold surface of the grass.	2	1 mark for water vapour condensing into water 1 mark for mentioning the cold surfaces
3(c)	There is snow on the top of Mount Everest because of the low temperature.	1	
	A thermal power station uses water to produce electricity.	1	
	Water freezes and ice melts at the same temperature.	1	
3(d)	Salt	1	
3(e)(i)	thermometer	1	
3(e)(ii)	Seawater boils at a higher temperature than water.	1	Accept the converse or they do not have the same boiling temperature/point

3(e)(iii)	To ensure that her results are reliable. OR So that she is sure/ is confident that her results are correct.	1	AW
Total marks		12	
4(a)(i)	nitrogen	1	
4(a)(ii)	water vapour	1	
4(a)(iii)	oxygen	1	
4(b)	21%	1	
4(c)(i)	Air pollution is when the air around us gets toxic with harmful gases or tiny particles.	1	AW
4(c)(ii)	Vehicles/ factories	1	Award a mark for a source shown in the diagram only
4(c)(iii)	Regular servicing of vehicles	1	Accept other correct ideas
	Use renewable and non-polluting sources of energy in factories.		
	Reduce the number of vehicles by using public transport or car pooling.		
4(c)(iv)	So as to prevent respiratory diseases/ skin diseases.	1	AW/ Accept other correct ideas
4(d)	By blocking their pores (thus preventing gas exchange)	1	
	By covering the surface with a layer of dust (thus preventing them from capturing enough sunlight)	1	
Total marks		10	
5(a)(i)	is transparent	1	
5(a)(ii)	is strong	1	
5(a)(iii)	is rare	1	
5(a)(iv)	absorbs sweat	1	
5(b)	Natural materials: Iron and wool	2	
	Man-made materials: Nylon and steel	2	
5(c)(i)	Plastic, Paper, Cotton cloth	1	All three should be in the correct order for the mark to be awarded
5(c)(ii)	Because the water has been absorbed by the cotton cloth.	1	Accept: The water may have spilled.
Total marks		10	
6(a)	Any herbivore: eg., cow, deer	1	
	Meat/ flesh/ chicken	1	
	Omnivore and any example of omnivore eg., humans, monkeys	2	
6(b)	There are 20 teeth in a set of milk teeth but 32 teeth in a set of permanent teeth	2	1 mark for the characteristic given for milk / permanent teeth 1 mark for the comparison
	Milk teeth are smaller than permanent teeth.		
	The roots of milk teeth are narrow whereas the roots of permanent teeth are wider.		
6(c)	By protecting their habitats	1+1 = 2	Any two correct answers
	By removing predators from their habitats		
	By conservation campaign restricting access and preventing people from collecting their eggs		

	By captive breeding		
6(d)	They could have gone extinct.	1	AW
6(e)(i)	1990 - Bar drawn at 60 2000 - Bar drawn at 120 2010 - Bar drawn at 150 2020 - Bar drawn at 210	1+1 = 2	Award one mark for any two bars correctly drawn
6(e)(ii)	60	1	
Total marks		12	
7(a)	B: In deserts	1	
	They have small leaves as spine (to prevent water loss)/ Their stem retains water. / They have deep roots.	1	
	A: In fresh water	1	
	Their roots grow into the soil at the bottom of ponds/lakes.	1	
	C: On trunks of trees	1	
	They live in damp places or in the shade./ They need minimal sunlight.	1	
7(b)	photosynthesis	1	
7(c)(i)	light/ sunlight	1	AW
7(c)(ii)	In order to compare the results.	1	AW
7(d)	Carbon dioxide/ water/ chlorophyll	2	Any two correct answers
Total marks		11	
8(a)	The Sun is a star and the Earth is a planet	1	Any one correct answer.
	The Sun is much bigger than the Earth		
8(b)(i)	24 hours	1	
	Movement of the Earth around the Sun once	1	
8(b)(ii)	The extra 1/4 day adds up to a full day every 4 years.	1	AW
8(c)(i)	Correct relative size and distance of the Sun, Earth and moon	1+1 = 2	1 mark for correct size 1 mark for correct distance from each other
8(c)(ii)	Shading of the side of the Earth away from the Sun	1	
8(d)	There is water, air and soil on the planet.	2	1 mark for each correct reason
	The Earth is at the right distance from the Sun so that it is neither too hot nor too cold.		AW
Total marks		9	
9(a)	B	1	
9(b)	The abnormal increase in the average temperature of the Earth's atmosphere	1	
9(c)	Burning of fossil fuel	1+1 = 2	Any two correct ideas

	Deforestation		
9(d)	By flooding of the coastal areas/ Increased coastal erosion	1+1 = 2	Any two correct ideas
	By disrupting the lagoon ecosystem/ By deepening of the lagoons		
9(e)	Solar water heaters do not produce carbon dioxide.	1	AW
9(f)	travelling to school: walk instead of using the car or bus/ car pooling/ use public transport instead of individual cars	1	Any correct idea
	in school: switch off the lights/fans when not in use	1	
Total marks		9	
10(a)(i)	X: A cell	1	Accep: battery
	Y: A (connecting electrical) wire	1	
10(a)(ii)	Because a complete electric circuit is formed/ because the circuit is closed.	1	AW
10(b)(i)	She should connect the positive terminal of one cell to the negative terminal of the other cell.	1	
	She should close the switch.	1	
10(b)(ii)	By using one more dry cell.	1	
10(c)	Because plastic is an insulator (and will prevent her from getting electric shock).	1	
10(d)	Chemical energy	1	
	Electrical energy	1	
	Sound energy	1	
Total marks		10	